

Kabir Rohit Khanna

Education

- Sep 2025 – Present **University of Geneva** *Geneva, Switzerland*
PhD in Physics | **Advisor:** Romain Vasseur
- Sep 2023 – Sep 2025 **University of Massachusetts** *Amherst, Massachusetts*
PhD in Physics | **Advisor:** Romain Vasseur
Relocated to University of Geneva with supervisor.
- Oct 2022 - Jun 2023 **University of Oxford** *Wadham College, Oxford, UK*
MSc. Mathematical and Theoretical Physics
Overall grade: Distinction
Thesis: Long Range Entanglement via Anomalies From Measurements
- Aug 2017 – May 2022 **Indian Institute of Technology Madras** *Chennai, Tamil Nadu*
Dual Degree - B.Tech in Engg. Design + M.Tech in Quantum Science & Technology
CGPA: 9.41/10 | **Rank:** 3/57
Thesis: Aspects of Quantum Information in Curved Spacetimes

Publications

- [1] **Kabir Khanna** and Romain Vasseur, *Measurement-Induced Entanglement in Conformal Field Theory*, **Phys. Rev. Lett.** **136**, 160402 (2026). [\[DOI\]](#) [\[arXiv\]](#)
- [2] **Kabir Khanna**, Abhishek Kumar, Romain Vasseur, and Andreas W. W. Ludwig, *Random Quantum Circuits with Time-Reversal Symmetry*, **Phys. Rev. Research** **8**, 013165 (2026). [\[DOI\]](#) [\[arXiv\]](#)

Preprints

- [3] **Kabir Khanna**, Sara Murciano, and Romain Vasseur, *Theory of Measurement-Altered Criticality*, arXiv:2608.05289 (2026). [\[arXiv\]](#)
- [4] Dolly Nambi, **Kabir Khanna**, Andrew Allocca, Thomas Iadecola, Ciarán Hickey, Romain Vasseur, and Justin H. Wilson, *Post-selected Criticality in Measurement-induced Phase Transitions*, arXiv:2603.15744 (2026). [\[arXiv\]](#)
- [5] **Kabir Khanna** and Romain Vasseur, *Universal Statistics of Measurement-Induced Entanglement in Tomonaga-Luttinger Liquids*, arXiv:2512.13809 (2025). [\[arXiv\]](#)
- [6] Raghvendra Singh, **Kabir Khanna**, and Dawood Kothawala, *Decoherence due to Spacetime Curvature*, arXiv:2302.09038 (2023). [\[arXiv\]](#)

““

Invited/Conference Talks and Lectures

- [1] QLANDS summer school on Non-Equilibrium, Circuits, and Complexity, June. 2026, *Exercise sessions on statistical mechanics of random quantum circuits.*
- [2] Swiss Quantum Days, Jan. 2026, *Measurement-Induced Entanglement in Gapless Quantum Matter.*
- [3] HRI Allahabad, Aug. 2025, *Lectures on Random Quantum Circuits and Their Statistical Mechanics.*
- [4] HRI Allahabad, Aug. 2025, *Measurement-Induced Entanglement in Conformal Field Theory.*

““

Awards and Recognitions

- 2023 Awarded the Graduate Fellowship worth \$5000 by UMass Amherst.
- 2022 OCSI Scholarship awarded by the Oxford Cambridge Society of India.
- 2021 Awarded[§] the DAAD WISE Research Scholarship.
- 2019 Awarded partial funding to attend the Wolfram Summer School at Waltham, Massachusetts.
- 2018 Awarded the best volunteer at NSS IITM.
- 2017 Silver medal for obtaining a top 4000 rank in JEE Advanced 2017 out of 1.2 million students.
- 2015 Awarded the best outgoing student from school, DPS Secunderabad.

Teaching and Volunteering

- Feb 2026 - May 2026 **Teaching Assistant for Theory Modules** *University of Geneva*
- Feb 2024 - May 2024 **Teaching Assistant for Astronomy 105 - Weather** *UMass Amherst*
- Sept 2023 - Dec 2023 **Teaching Assistant for Phys 118 - Weather and Our Atmosphere** *UMass Amherst*
- Jan 2022 - May 2022 **Teaching Assistant for PH3520 Quantum Physics** *IIT Madras, Chennai*
- May 2018 - Jul 2018 **Mathematics and Physics Tutor** *Lamdon School, Leh*

[§]Did not accept.